conviction_tilt_max_legacy

equity strategy | family: equity-max | status: reference | derived from `conviction_tilt_max`

What this strategy does

conviction_tilt with bigger bets for more return: base unit 0.007, 6x leverage cap -- ORIGINAL leverage cap (new positions only, open book ignored)

Why it should work. Concentrate capital where the surprise is largest; neutralize size so it is not a hidden bet.

Design notes (from the strategy's source file).

Max-size variants ("<name>_max") of the hand-built equity strategies.

Same signal, trim, tilt and neutralization as each base strategy; only the two dials that set how big and how levered the book is change (base_unit_fraction, gross-leverage cap = 6x). The multiplier for each was picked by sweeping 1.5x..12x the original base unit and taking the highest return subject to Sharpe >= 0.6 and max drawdown >= -40% (a worse drawdown is a fund-ending event, not a risk preference). The untouched 5%-of-ADV liquidity cap remains the real capacity constraint, and bites roughly four times as often at this sizing.

These are generated from the base strategy classes, so a change to a base strategy's weights flows through automatically.

parameter	value
base_unit_fraction	0.007
max_gross_leverage	6.0
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	entry_cohort
beta_overlay	0.0

Performance

metric	conviction tilt max legacy	market (CRSP VW)
Annualized return	10.70%	9.94%
Annualized volatility	12.25%	19.35%
Sharpe (NAV-pct)	0.89	0.59
Sortino	1.27	
Max drawdown	-40.0%	-55.5%
Calmar	0.27	
Final NAV ($10M start)	$72,371,948	
Worst day	-6.54%	
% positive days	51.9%	

Sharpe 1996-2001	1.30	
Sharpe 2002-2007	1.54	
Sharpe 2008-2013	0.07	

Correlation of daily returns with the market: **0.03**.

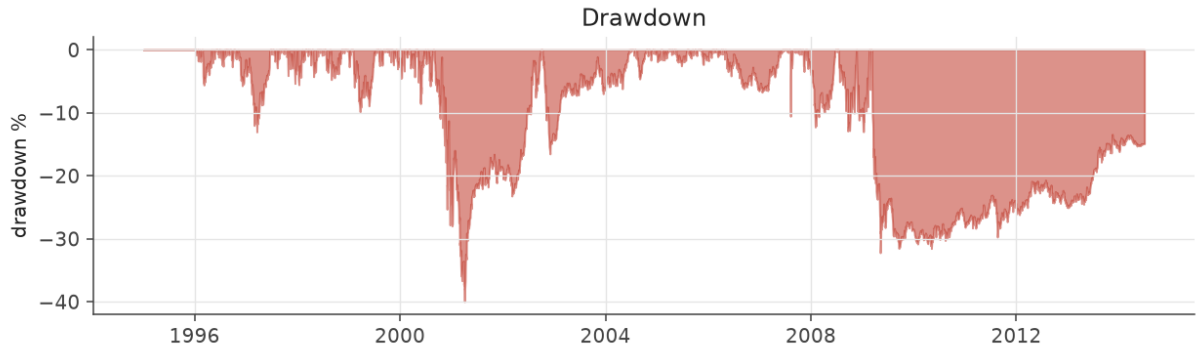
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 0.891

• best return among equity: `market_boost_max` (0.116); this one: 0.107

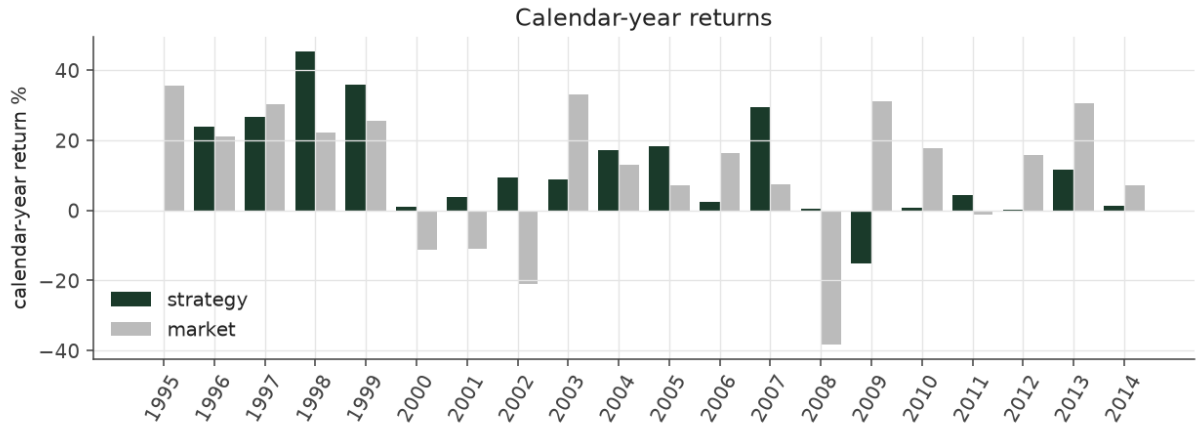
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.267



NAV (log scale)



Drawdown from running peak

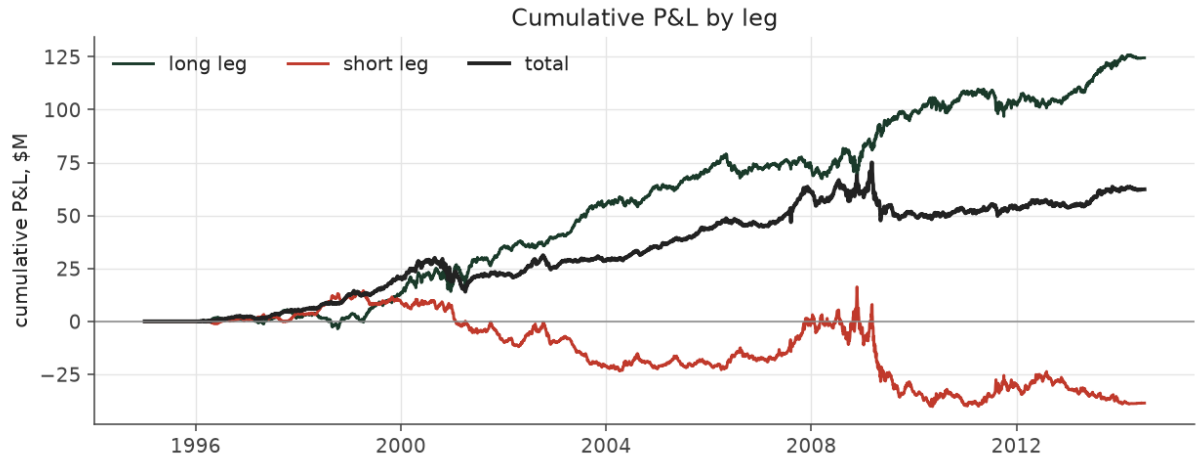


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	124.48	-38.59
P&L after costs ($M)	113.18	-50.81
of which common (average-event) component ($M)	105.53	-105.53

of which excess / PEAD component ($M)	17.78	67.56
excess minus costs ($M)	6.51	55.37
gross return per $ traded (bp)	189.32	-56.80
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	0.95	-0.15

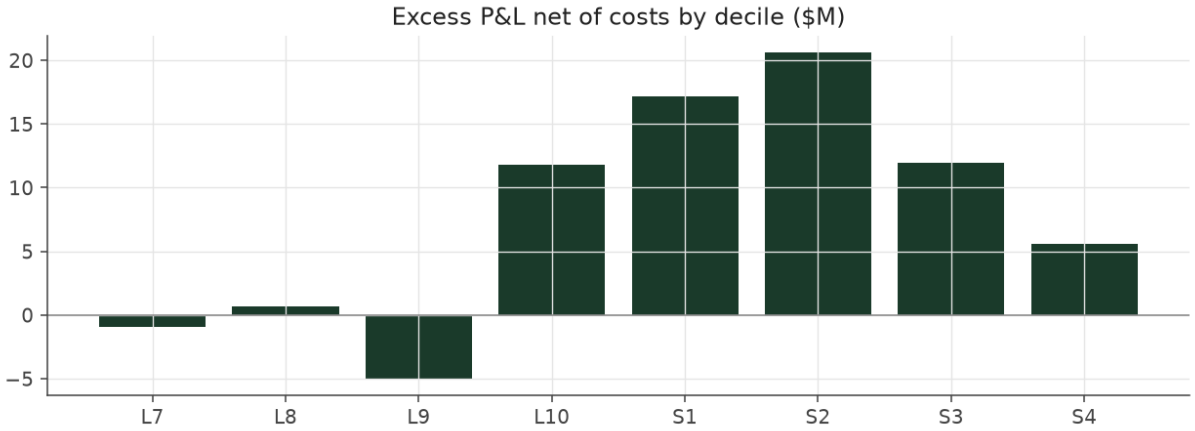
Daily P&L correlation between the two legs: **-0.72**.

• **The short leg loses money in absolute terms** (-38.6M before costs) while the long leg earns +124.5M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 105.5M / 105.5M. Stripping that common component, the surprise signal itself earned +17.8M on the long side and +67.6M on the short side.

• The two legs' daily P&L are -0.72 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 0.95 long / -0.15 short vs 0.89 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,029	531.28	8.56	0.00	0.97	-0.97	-1.6%	-18.21
long	8	16,588	1,498.23	26.40	3.34	2.65	0.68	1.1%	4.57
long	9	17,747	2,072.79	26.15	-1.52	3.50	-5.02	-8.1%	-24.23
long	10	17,174	2,472.81	52.07	15.96	4.17	11.79	19.1%	47.69
short	1	17,169	2,422.99	-18.52	21.57	4.38	17.19	27.8%	70.94
short	2	17,281	2,130.37	-14.14	24.39	3.77	20.62	33.3%	96.78
short	3	15,970	1,602.41	-13.84	14.90	2.92	11.98	19.4%	74.78
short	4	9,171	638.34	-4.31	6.70	1.15	5.55	9.0%	86.95



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,213	602.96	4.85	8.89	2.02	6.87	11.1%	113.97
2	22,885	2,160.87	17.58	23.66	5.48	18.18	29.4%	84.14

3	25,501	3,273.89	21.32	25.52	6.45	19.07	30.8%	58.26
4	27,583	3,629.42	11.28	15.96	5.50	10.45	16.9%	28.80
5	28,947	3,702.08	7.34	11.31	4.06	7.25	11.7%	19.58

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,650	301.82	5.97	8.06	1.51	6.55	10.6%	217.01
2	23,084	1,899.67	12.57	20.66	5.70	14.96	24.2%	78.73
3	25,851	3,614.33	23.26	31.74	7.23	24.51	39.6%	67.81
4	27,665	3,809.07	10.53	15.10	5.33	9.76	15.8%	25.63
5	28,874	3,743.30	10.05	9.77	3.74	6.03	9.7%	16.11

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,855	5,376.99	-6.32	24.12	8.71	15.41	24.9%	28.65
41-60d	25,652	2,972.62	15.99	14.05	5.33	8.72	14.1%	29.32
61-90d	35,324	3,914.33	19.92	19.75	7.17	12.58	20.3%	32.14
>90d	12,298	1,105.28	32.79	27.43	2.30	25.12	40.6%	227.32

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,895	2,125.65	21.61	24.00	3.67	20.33	32.9%	95.65
2002-07	39,956	4,622.80	35.24	41.70	8.03	33.67	54.4%	72.84
2008-13	45,278	6,620.77	5.52	19.64	11.81	7.82	12.6%	11.82

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	22,094	2,356.09	15.94	8.48	4.12	4.36	7.0%	18.50
Chems	2,590	305.15	1.28	2.16	0.51	1.65	2.7%	53.97
Durbl	2,843	306.15	0.81	2.30	0.55	1.75	2.8%	57.23
Enrgy	6,194	728.42	8.18	10.69	1.19	9.50	15.4%	130.41
Hlth	11,696	1,213.37	10.05	8.11	2.27	5.84	9.4%	48.16
Manuf	12,034	1,336.21	2.73	6.91	2.32	4.59	7.4%	34.36
Money	17,583	2,118.09	7.17	19.55	3.69	15.86	25.6%	74.88
NoDur	5,081	574.32	2.52	2.88	0.98	1.90	3.1%	33.13
Other	18,377	2,048.37	7.91	12.37	3.86	8.51	13.8%	41.57
Shops	11,984	1,335.59	7.65	8.85	2.35	6.50	10.5%	48.68

Telcm	3,981	448.63	-1.34	1.27	0.73	0.53	0.9%	11.87
Utils	4,672	598.84	-0.53	1.77	0.95	0.82	1.3%	13.74

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,523	4,710.15	10.53	19.95	7.61	12.34	19.9%	26.20
low vol	40,091	4,230.51	24.63	32.51	8.07	24.43	39.5%	57.75
mid vol	37,903	4,264.38	26.18	32.48	7.52	24.97	40.3%	58.55

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	10.70%	0.00%
0.5%	6.7	10.15%	0.55%
1.0%	13.3	9.55%	1.15%
2.0%	26.6	8.12%	2.58%
5.0%	66.6	-2.80%	13.50%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.